

Byeol Kim Lux

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Last updated: August 2026

MAJOR RESEARCH INTEREST

- Neurobiological and physiological mechanisms of emotion and subjective experience
- Brain-body interactions across neural, autonomic, and immune systems
- Psychoneuroimmunological pathways linking stress, emotion, and long-term health outcomes

EDUCATION & TRAINING

Ph.D. candidate in **Cognitive Neuroscience** Sep 2021 - Present

Department of Psychological and Brain Sciences,
Dartmouth College, Hanover, NH, USA
Lab: Cognitive and Affective Neuroscience Laboratory ([CANLab](#))
(PI: Dr. Tor D. Wager)

Post-Master Research Assistant Sep 2019 – Aug 2021

Center for Neuroscience Imaging Research (CNIR),
Institute of Basic Science (IBS), Suwon, South Korea
Lab: Computational Cognitive Affective Neuroscience Laboratory ([COCOAN Lab](#))
(PI: Dr. Choong-Wan Woo)

M.S. in **Biomedical Engineering**, Sep 2017 – Aug 2019

Department of Biomedical Engineering,
Sungkyunkwan University, Suwon, South Korea (Advisor: Dr. Choong-Wan Woo)
Thesis: [*Understanding the brain representations of endogenous affective thoughts.*](#)

B.S. in **Astronomy**, B.A. in **Psychology**, Mar 2012 – Aug 2017

Department of Astronomy & Department of Psychology
Yonsei University, Seoul, South Korea

HONORS and AWARDS

Trainee Professional Development Award (November 2026)

Society for Neuroscience

Wisconsin Symposium on Emotion Travel Award (April 2026)

University of Wisconsin-Madison

Neukom Travel Award (Jan 2026)

The Neukom Institute for Computational Science, Dartmouth College

Guarini Travel Award (Mar 2024)

Guarini School of Graduate and Advanced Studies, Dartmouth College

KIST & AKN Outstanding research award (Nov 2022)

Association of Korean Neuroscientists, South Korea

Fulbright graduate student program award (2021 – 2023)

Bureau of Educational and Cultural Affairs, United States Department of State

Best presentation award (Dec 2018)

Center for Neuroscience Imaging Research, Institute for Basic Science, South Korea

Shimsan scholarship for advanced graduate students (Fall 2018)

Sungkyunkwan University, South Korea

Best presentation award (May 2018)

Korean Society for Cognitive Science, South Korea

High academic achievement scholarship (Fall 2015, Spring 2016, 2013)

Yonsei University, Seoul, South Korea

Scholarship for undergraduate students (Fall 2012)

Sinchon regional support association, Seoul, South Korea

PUBLICATIONS

Published under the name Byeol Kim (2021-2022) and Byeol Kim Lux (2023-),

*Co-first authors, ✉ Co-corresponding authors

In preparation & Under review

Lux, B. K., Kos, M. C., Ward, D., Kolling IV, F. W., Li, X., Jakubzick, C. V., & Wager, T. D. Vicarious trauma primes innate immunity and reconfigures human brain networks. [bioRxiv](#) (*Under review in Nature Communications*)

Han, J.*, **Lux, B. K.***, Lee, E., Yoo, Y. S. & Woo, C.-W. Modeling spontaneous thought: A network- and language-based computational method. (*Under review in IEEE Transactions on Affective Computing*)

Lux, B. K.*, Shin, H.*, Kim, H. J. & Woo, C.-W. Bodily maps of spontaneous thought. [bioRxiv](#) (*Submitted*)

Published

Lux, B. K.*, ✉, Lee, E.*, Han, J., Lee, S.-H., Gim, S., Choi, I., & Woo, C.-W. ✉ Spontaneous thought dynamics as a signature of positive and negative affectivity. [Communications Psychology](#). (2026, in press)

Shin, W. G., Ahn, J., Choe, K. W., Lee, H., Han, J., Lee, E., **Lux, B. K.**, Woo, C.-W., Choi, I., & Sul, S. (2025). Happier individuals generate more spontaneous thoughts about friends and value relationships over money. *Communications Psychology*. DOI: [10.1038/s44271-025-00341-3](#)

Kim, H. J., **Lux, B. K.**, Lee, E., Finn, E., & Woo, C. -W. (2024). Brain decoding of spontaneous thought: Predictive modeling of self-relevance and valence using personal narratives. *Proceedings of the National Academy of Sciences*. DOI: [10.1073/pnas.2401959121](#)

Jang, S., Lee, K. R., Goh, G., Kim, D., Yun, G., Kim, N., **Lux, B. K.**, Woo, C.-W, Kim, H., & Park, Y. W. (2024). Design and field trial of EmotionFrame: exploring self-journaling experiences in homes for archiving personal feelings about daily events. *Human-Computer Interaction*. DOI: [10.1080/07370024.2023.2219259](#)

Kim, J., Andrews-Hanna, J. R., Eisenbarth, H., **Lux, B. K.**, Kim, H., Lee, E., Lindquist, M., Losin, E. A. R., Wager, T. D., & Woo, C.-W. (2023). A Dorsomedial Prefrontal Cortex-based Dynamic Functional Connectivity Model of Rumination. *Nature Communications*. DOI: [10.1038/s41467-023-39142-9](#)

Lux, B. K., Andrews-Hanna, J. R., Han, J., Lee, E., & Woo, C.-W. (2022). When self comes to a wandering mind: Brain representations and affective dynamics of spontaneous thought. *Science Advances*. DOI: [10.1126/sciadv.abn8616](#)

Andrews-Hanna, J. R., Woo, C.-W., Wilcox, R., Eisenbarth, H., **Kim, B.**, Han, J., Losin, E. A. R., & Wager, T. D. (2022). The conceptual building blocks of everyday thought: Tracking the emergence and dynamics of ruminative and non-ruminative thinking. *Journal of Experimental Psychology: General*. DOI: [10.1037/xge0001096](https://doi.org/10.1037/xge0001096) | [PDF](#)

CONFERENCE PRESENTATIONS (First author only)

- Lux, B. K.**, Kos, M., Ward, D., Li, X., Kolling, F., Jakubzick, C., Wager, T.D. (July 2026).
Morally Salient Vicarious Trauma Enhances Insula-Executive and Sensorimotor-Executive Connectivity and Tracks Personal Distress in Healthy Adults
Poster presentation at the annual meeting of Computational Psychiatry, New Haven, CT, USA.
- Lux, B. K.**, Kos, M., Ward, D., Li, X., Kolling, F., Jakubzick, C., Wager, T.D. (May 2026).
Vicarious Trauma Primes Innate Immunity and Reconfigures Human Brain Networks
Poster presentation at the annual meeting of Society of Biological Psychiatry, New York City, USA.
- Lux, B. K.**, Kos, M., Ward, D., Li, X., Kolling, F., Jakubzick, C., Wager, T.D. (April 2026).
The Neural Cost of Witnessing Suffering: Network Disruption and Sustained Arousal After Vicarious Trauma
Poster presentation at the annual meeting of Wisconsin Symposium on Emotion, Madison, USA.
- Lux, B. K.**, Kos, M., Ward, D., Li, X., Kolling, F., Jakubzick, C., Wager, T.D. (Nov 2025).
Systemic impacts of vicarious trauma induced by documentary footage of animal suffering: insights from fMRI, physiology, and monocyte gene expression
Poster presentation at the annual meeting of the Society for Neuroscience, San Diego, USA.
- Lux, B. K.**, Kos, M., Ward, D., Jakubzick, C., Wager, T.D. (June 2024)
Physiological, Immune, and Neural Responses to Witnessing Animal Suffering and Connection
Poster presentation at the annual meeting of the Organization for Human Brain Mapping, Seoul, South Korea.
- Lux, B. K.**, Kos, M., Ward, D., Wager, T.D. (April 2024)
Exploring the Psychological and Neurobiological Impact of Animal Suffering on Humans
Poster presentation at the annual meeting of the Social and Affective Neuroscience Society, Toronto, Canada.
- Lux, B. K.**, Kragel, P. A., Čeko, M., Theriault, J., Chen, D., Satpute, A. B., Wald, L. L., Lindquist, M. A., Barrett, L. F., & Wager, T. D. (Nov 2022).
Identification of a human parabrachial-amygdala pathway with 7T fMRI
Poster presentation at the annual meeting of the Society for Neuroscience, San Diego, USA.
- Kim, B.**, Andrews-Hanna, J. R., Han, J., Lee, E., & Woo, C.-W. (April 2021).
When self comes to a wandering mind: Brain representations and affective dynamics of spontaneous thought.
Poster presentation at the annual meeting of the Social & Affective Neuroscience Society, held virtually.
- Kim, B.**, Andrews-Hanna, J. R., & Woo, C.-W. (Oct 2019).
Understanding the brain representations of endogenous affective thoughts.
Poster presentation at the annual meeting of the Society for Neuroscience, Chicago, USA.
- Kim, B.**, Andrews-Hanna, J. R., & Woo, C.-W. (Oct 2019).
Understanding the brain representations of endogenous affective thoughts.

- Poster presentation at the Institute of Basic Science Conference on Neuroimaging, South Korea.
Kim, B., Oh, J., Andrews-Hanna, J. R., & Woo, C.-W. (Nov 2018).
Dynamic modeling and brain decoding of internal thoughts and emotions.
Poster presentation at the annual meeting of the Society for Neuroscience, San Diego, USA.
Kim, B., Andrews-Hanna, J. R., & Woo, C.-W. (Oct 2018).
Dynamic modeling and brain decoding of internal thoughts and emotions.
Oral presentation at the monthly meeting of Center for Neuroscience Imaging Research, South Korea.
Kim, B., Jung, M., Park, J., Andrews-Hanna, J. R., & Woo, C.-W. (May 2018).
Dynamic modeling and brain decoding of internal thoughts and emotions.
Poster presentation at the annual meeting of the Korean Society for Cognitive Science,
South Korea.

TEACHING EXPERIENCE & GUEST LECTURE

- Guest lecturer at Mind, Brain, and Health at Dartmouth College
(Title: Seeing harm, preparing for injury: anticipatory immune programming and neural network reconfiguration to vicarious trauma, Fall 2025)
Teaching Assistant for Learning at Dartmouth College (Spring 2024)
Teaching Assistant for Lab in Psychological Science at Dartmouth College (Summer 2023)
Teaching Assistant for Systems Neuroscience with Laboratory at Dartmouth College (Spring 2023)
Teaching Assistant for Principles of Human Brain Mapping with fMRI at Dartmouth College (Winter 2023)
Guest lecturer at Advanced Affective Neuroscience (Graduate course) at Sungkyunkwan University
(Title: Research trends of self in affective neuroscience, Fall 2020)
Undergraduate student research mentor of 2020 summer coding & research project
at COCOAN Lab, Sungkyunkwan University (Title: Maps of bodily sensation for subjective feeling from free-associated concept, Summer 2020)
Undergraduate student research mentor of web-based FAST (Free association semantic task) project at
COCOAN Lab, Sungkyunkwan University. (Fall 2019-Summer 2021)
Guest lecturer at Affective Neuroscience (Undergraduate course) at Sungkyunkwan University. (Title: Finding self, free association semantic task, Fall 2019)
Undergraduate student research mentor of the Summer internship program
at Center for Neuroscience Imaging Research, Institute of Basic Science.
(Title: Grid-like neural representation of navigating affective space: Evidence for the existence of an affective map?, Summer 2019)
Undergraduate student research mentor of the C-School undergraduate research project
at Sungkyunkwan University. (Title: Neural representations of valence and self-relevance in self-generated thoughts, Fall 2017)
Mathematics instructor for middle school students at the Aporia Mathematics Education Institute, Seoul,
South Korea (2015-2017)
Personal mentoring mathematics (from elementary school to high school level) and science (from middle school to high school level) for adolescents between the ages of 7 and 18, Seoul, South Korea
(2011-2016)

PATENT APPLICATION

- Woo, C.-W., Lee, E., & **Kim, B.** (2026). "Method and Apparatus for Training Machine-Learning Model for Predicting Affectivity." South Korea Patent No. 10-2939550, filed July 05, 2022, issued March 11, 2026.
- Woo, C.-W., **Kim, B.**, & Han, J. (2022). "Visualization and Emotion-Predictive Modeling of the Stream of Thought." South Korea Patent No. 10-2441591, filed July 09, 2020, issued September 02, 2022.

OTHER ACADEMIC EXPERIENCES

Santa Fe Institute Complex System Summer School (June-July 2026)
International Graduate Mentorship Program, Dartmouth College (2023-26)
Grad Student Virtual Professional Development Bootcamp, The Leadership Alliance (Jan 2025)
Prospective Ph.D. & RA Event in Psychology (PPREP), Harvard University (Oct 2020)
Member of the student council of Department of Biomedical Engineering, Sungkyunkwan University (Spring 2018)
Member of the student council of Department of Astronomy, Yonsei University (Spring 2014)

SKILLS

Programming Language	MATLAB, Python, Unix, HTML, JavaScript
Neuroimaging	SPM, CANlab tools (Dr. Tor Wager's lab tools), FSL, fMRIPrep, tedana, afni, GIFT
Statistics	R, SPSS, Jamovi
Experiment Software	Psychtoolbox, PsychoPy, Pavlovía